DAY -45

Daily Learning Diary

Date: March 5, 2026

Topic: Model Evaluation (House Price Prediction)

I. Overview

Today, I focused on measuring how well my machine learning model performs. After building a model to predict house prices, I needed to know if the predictions were actually close to the real market values. I used two main metrics: **Mean Squared Error (MSE)** and **R2 Score**.

II. Key Concepts

1. Mean Squared Error (MSE)

MSE tells me the average "error" of my predictions.

- **How it works:** It takes the difference between the actual price and the predicted price, squares it (to remove negative signs), and then finds the average.
- **What it means:** A lower MSE is better. It means the "distance" between my guesses and the truth is small.
- **Formula:**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

2. R2 Score (Coefficient of Determination)

R2 tells me how much of the variation in house prices is actually explained by my model.

- **The Scale:** It usually ranges from **0 to 1**.
  - **0:** The model is no better than just guessing the average price every time.
  - **1:** The model is perfect and hits every price exactly.
- **What it means:** If my R2 is 0.85, it means 85% of the price changes can be explained by the features I used (like square footage or location).

III. Implementation: House Price Dataset

Implementation Workflow

1. **Data Preparation & Stratified Split** I used a **stratified split** to ensure the training and testing sets have the same percentage of fraud cases as the original data. This is crucial for imbalanced datasets.
2. **Feature Scaling (Standardization)** SVMs are sensitive to the scale of data. I used StandardScaler to make sure all features have a mean of 0 and a variance of 1, preventing one feature from dominating the model.
3. **Support Vector Classifier (SVC)** I initialized the SVM model. Its job is to find the best boundary (hyperplane) that separates fraudulent transactions from legitimate ones.
4. **Hyperparameter Tuning with GridSearchCV** Instead of guessing the best settings, I used **Grid Search** with 3-fold cross-validation to test combinations of:
    - **C:** Controls the trade-off between a smooth boundary and classifying training points correctly.
    - **Kernel:** Tested linear and rbf to see if the data is better separated by a straight line or a non-linear curve.

IV. Reflections

- **MSE** is great for math, but because the errors are squared, the number can get very large. It’s hard to "visualize" in dollars.
- **R2 Score** is much easier to explain to others because it gives a clear picture of how "smart" the model is relative to a simple average.